

## Workshop 2 Web Scraping

[Click here to access the practice code](https://tinyurl.com/4zsymu7f)

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### Background Information

#### 1. What is web scraping?

Web scraping is the automated process of extracting data from a website. Other expressions include web harvesting, data extraction, web data extraction, data scraping and data mining.

#### 2. What is scraping data?

Scraping data is about getting juicy unstructured data and transforming it into something more useful.

#### 3. What are the benefits of web scraping?

Fast and efficient, data extraction at scale, cost-effective and flexible, reliable and robust performance, low maintenance costs, delivers structured data, etc.

#### 4. What is web scraping used for? (Some real-world use cases)

Market research, healthcare data extraction, sentiment analysis, academic research, etc.

### Python Basics Scaffold

#### 1. Comments

`##` → This indicates a comment in Colab/Python.

Comments are not executed. They are just notes for humans.

#### 2. Installing Packages

```
!pip install youtube-comment-downloader
```

`!` → In Colab, this allows you to run a system command (like pip install).

`pip` → A tool to install Python packages.

### 3. Importing Libraries

```
import pandas as pd
```

import → Bring a tool/library into your code.

pandas → A data processing library.

as pd → Create a short nickname (alias) for easier typing.

### 4. Variables and Assignment

= → Assignment operator (store something in a variable).

Example: x = 5

### 5. Lists

[ → Start a list.

] → End a list.

, → Separate items inside the list.

Example: urls = ['link1', 'link2']

### 6. Strings (Text)

'text' or "text" → Quotation marks mean the content is text (a string).

Python treats text differently from numbers.

### 7. f-Strings

```
name = "Anna"  
f"Hello {name}"
```

f"...{variable}..."

Formatted string literal: allows you to embed variables directly inside a string.

### 8. Loop Index

```
json_file = f"YoutubeComments_{i}.json"
```

i → Often used as a counter (0, 1, 2...).

Used in loops to track position.

### 9. Modules and Functions

Structure: module\_name.function\_name(parameter)

Example: `os.remove('data.json')`

`os` → A built-in Python module for interacting with the operating system.

`( )` → Used to call a function.

## 10. File Encoding and Modes

```
with open(json_file, "r", encoding="utf-8") as f
```

`'r'` → File open mode: read-only.

`utf-8` → A text encoding standard (supports many languages).

## 11. DataFrame (Pandas)

```
each_url_data = pd.DataFrame(json_data)
```

`DataFrame` → A table-like data structure (like Excel).

## 12. Creating a New Column

```
each_url_data['video_url'] = url
```

`each_url_data` → A variable name (a DataFrame).

`['video_url']` → Pandas syntax to create or access a column.

`= url` → Assign the value of `url` to that column.

## 13. Index Handling

Python starts counting from 0, not 1.

`ignore_index = True` → Reset row numbers when combining tables.

`index=False` → Do not save row numbers when exporting to CSV.